

Bahman Gharesifard
Curriculum vitae: September, 2026

Professor

Department of Mathematics and Statistics
Queen's University

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POSITIONS

Professor of Mathematics (with tenure), Queen's University July 2024 –
Department of Mathematics and Statistics

Professor (with tenure), University of California, Los Angeles July 2021– July 2024
Area Director for Signals and Systems July 2023– July 2024
UCLA Samueli School of Engineering
Department of Electrical and Computer Engineering

Associate Professor, Queen's University July 2019 – June 2021
Department of Mathematics and Statistics

Alexander von Humboldt Fellow, University of Stuttgart October 2019 – December 2020
Institute for Systems Theory and Automatic Control

Ontario/Baden-Württemberg Visiting Faculty, University of Stuttgart Summer 2016
Institute for Systems Theory and Automatic Control

Assistant Professor, Queen's University July 2013 – July 2019
Department of Mathematics and Statistics

Postdoctoral Research Associate, University of Illinois, Urbana-Champaign March 2012 – July 2013
Coordinated Science Laboratory
Department of Electrical and Computer Engineering

Postdoctoral Research Associate, University of California San Diego September 2009 – March 2012
Jacobs School of Engineering
Department of Mechanical and Aerospace Engineering

EDUCATION

Ph.D., Mathematics
Queen's University, Kingston, ON, Canada October 2009

B.Sc. and M.Sc., Mechanical Engineering (Dynamics and Control)
Shiraz University, Shiraz, Iran May 2005

RESEARCH INTERESTS

Decision Theory: systems and control; stability of discrete-time/continuous-time dynamical systems; networked control systems; learning for control; stochastic control; sensor networks; discontinuous dynamical systems; nonlinear and geometric control; autonomy; controllability of partial differential equations; economic dispatch of energy in networked power systems; identification and adaptation

Machine Learning and Optimization Theory: mathematical theory of neural networks, and transformers; sample complexity analysis; reinforcement learning; decentralized optimization; statistical learning; submodular optimization; online learning; GANs; model collapse in generative models

Optimal Transport: intersections with control, learning, game theory, and optimization; continuity equations

Social Networks: opinion dynamics; diffusion models; reinforcement schemes; random graphs

Game Theory and Economics: game theory and strategic interactions; hypergames; market design in distributed settings; learning in games

Stochastic Processes and Probability Theory: stochastic reinforcement processes, stochastic differential equations and control; rough path theory

Global Analysis: applied differential geometry; exterior differential systems; Lie theory, Riemannian geometry; geometric theory of partial differential equations

AWARDS AND HONORS

Invited Speaker , Mathematical Foundations of AI and Data-Driven Scientific Computing, National University of Singapore, Singapore	2026
Semi-plenary Speaker , 27th International Symposium on Mathematical Theory of Networks and Systems University of Waterloo	2026
The 2024 CSS TC on Robust and Complex Systems Outstanding Student Paper Prize	2024
Canadian Society of Information Theory Best Paper Award	2022
Best conference paper that has been accepted by either the Biennial Symposium on Communications (BSC) or the Canadian Workshop on Information Theory (CWIT)	
SIAG/CST Best SICON Paper Prize 2021	2021
Awarded to the author(s) of the two most outstanding papers, as determined by the prize committee, published in the SIAM Journal on Control and Optimization (SICON) in the three calendar years before the award year	
Booz Allen Hamilton Distinguished Colloquium Lecture , University of Maryland	2020
Alexander von Humboldt Fellowship for experienced researchers	2019
The 2019 CAIMS-PIMS Early Career Award	2019
Jointly awarded by the Canadian Applied & Industrial Math Society and the Pacific Institute for the Mathematical Sciences	
NSERC Discovery Accelerator Supplement	2019
supplemental funding with 125 recipients among over 3000 applicants in Canada	
Finalist for Best Paper Award (as a supervisor), American Control Conference	2017
Class of 2020 Engineering and Applied Science First Year Instructor Teaching Award	2017
Class of 2018 Engineering and Applied Science First Year Instructor Teaching Award	2015
Shortlisted for the Frank Knox Award for Excellence in Teaching	2014

RESEARCH GRANTS

NSERC Discovery Grant	2014-
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National Science Foundation

Structural Frameworks for Output Control of Continuum Ensemble Systems (with Xudong Chen) 2023

National Science Foundation

Formally correct deep perception for cyber-physical systems (with Paulo Tabuada) 2022

Fields Centre for Quantitative Analysis and Modeling

co-PI of one of the six labs 2018-2021

Alexander von Humboldt Fellowship for experienced researchers

2019-2020

Ontario/Baden-Württemberg Exchange Program

May-July 2016

visiting the Institute for Systems Theory and Automatic Control, University of Stuttgart

NSERC Discovery Accelerator Supplement

2019-2024

Senate Advisory Research Committee Grant

2014-2015

ADMINISTRATIVE ACTIVITIES

Colloquium co-chair, Queen's University, 2025-2026

Renewal, Promotion, and Tenure Committee, Queen's University, 2025-2026

Area Director for Signals and Systems, UCLA ECE 2023-2024

UCLA Samueli Faculty Executive Committee 2022-2024

Chair, Honors Committee, UCLA Samueli 2023-2024

Colloquium Chair, UCLA 2022

Signals and Systems Graduate Admissions Committee, UCLA 2022-

Chair, CAIMS-PIMS Early Career Award 2021-

CAIMS-PIMS Early Career Award Committee 2019-

Postdoctoral Committee, Queen's University 2017-2018

Appointment Core Committee, Queen's University 2015-2016

Chair, Advisory Committee on Policy Queen's University 2016-2017

Advisory Committee on Policy, Queen's University 2015-2017

First Year Engineering Curriculum Review Committee (FYCRC), Queen's University 2014-2015

Graduate Attribute Project Planning Committee (GAPPC), Queen's University 2014-2015

Appointment Core Committee, Queen's University 2014-2015

Interviewing for Academia and Mock Interview Practice, School of Graduate Studies 2014

Graduate committee, Queen's University 2014-2016

Math & Stats Colloquium Chair, Queen's University 2014-2015

Ontario Graduate Scholarship Reviewer, Queen's University 2014,2016,2018

Major Undergraduate Admission Award Reviewer, Queen's University 2013-2015,2018

Judge, Andrina McCulloch Public Speaking competition, 2014, 2015

Mathematics and Engineering Committee, Queen's University 2013-2019

REVIEW, EDITORIAL DUTIES, and ORGANIZED SESSIONS AND CONFERENCES**Vice Program Chair**

Control and Decision Conference

Hawaii, USA December 2026

Editorial Board

SIAM Advances in Design and Control

January 2026-

Session Chair

Başar Fest

Celebration of his 80th birthday of Tamer Başar

University of Illinois, Urbana-Champaign,

September 2025

Program Chair

Annual Canadian Applied and Industrial Mathematics Society
to be held in Kingston, ON, Canada,

June 2024

Program Committee Member

SIAM Conference on Control and Its Applications, Montreal, Canada

July 2025

Program Committee Member

IFAC Conference on Networked Systems, Hong Kong,

June 2025

Program Committee

24th ACM International Conference on Hybrid Systems: Computation and Control

July 2021

International Program Committee

SIAM Conference on Control and Its Applications, Philadelphia, US

July 2023

IEEE CDC 2023 Best Student Paper Award Committee

Control and Decision Conference, Singapore

2023

IEEE CDC 2023 Best Student Paper Award Committee

Control and Decision Conference, Milan, Italy

2024

Associate Editor:

IEEE Transactions on Control of Network Systems

2021-2025

Program Committee Member:

Learning for Dynamics and Control (L4DC),
UC Berkeley, ETH, and Stanford

2020-2022

Associate Editor

IEEE Control Systems Letters

2019-2022

Conference Editorial Board:

IEEE Control Systems Society

2016-2020

Scientific Committee Member:

International Young Researcher Workshop

Geometry, Mechanics, and Control

2015-

Locations: Coimbra (Portugal), Padova (Italy), Paris (France), Madrid (Spain)
Göttingen (Germany), KULeuven (Belgium), Verona (Italy)

Co-organizer (7th Biannual Meeting on System and Control Theory)

Fields Institute for Research in Mathematical Sciences Workshop

with A. Lewis, A.R. Mansouri, S. Yüksel (Queen's)

2016

Technical Program Committee:

55th Conference on Decision and Control, Las Vegas, USA

2016

Minisymposium co-organizer (Nonsmooth and variational techniques in control)

SIAM Conference on Control and its Applications, Paris, France

with R. Kipka (Queen's)

2015

Invited session organizer (Controllability of network control systems)
 Control and Decision Conference, Osaka, Japan
 with M-A. Belabbas (UIUC) and C. Aguilar (CSUB) 2015

Invited session organizer (Controllability of network control systems)
 Control and Decision Conference, Los Angeles
 with M-A. Belabbas (UIUC) and C. Aguilar (CSUB) 2014

Invited session organizer (Stabilization of network control systems)
 Control and Decision Conference, Los Angeles
 with M-A. Belabbas (UIUC) and C. Aguilar (CSUB) 2014

Minisymposium co-organizer (Adversarial networks: learning and information exploitation)
 SIAM Conference on Control and its Applications, Baltimore
 with J. Cortes (UCSD) 2011

Tutorials Chair:
 27th Biennial Symposium on Communications
 Kingston, Canada 2014

Referee for Grant Proposals: National Discovery Grants (Canada), National Science Foundation (USA),
 FNRS Fund for Theoretical Research (Belgian), Luxembourg National Research Fund (Luxembourg), Israel
 Science Foundation (Israel)

Reviewer for:
 Journal of Nonlinear Analysis, IEEE Transactions on Network Science and Engineering, IEEE Control
 Systems Letters, European Journal of Operational Research, Journal of Special Matrices, IEEE Transactions
 on Signal Processing, Games, SIAM Journal on Control and Optimization, IEEE Transactions on Systems,
 Man, and Cybernetics: Systems, IEEE Transactions on Automatic Control, IEEE Transactions on Control
 of Network Systems, IEEE Transactions on Power Systems, IEEE Transactions on Energy Conversion,
 Automatica, European Journal of Control, Journal of Geometric Mechanics, Mathematics of Control, Signals,
 and Systems, IEEE Journal of Robotics and Automation, Journal of Systems, Control and Communication,
 ASME Journal of Dynamic Systems, Measurement and Control, Nonlinear Dynamics, Computational and
 Applied Mathematics, IEEE Transactions on Control Systems Technology, IEEE Control Systems Magazine,
 Annual Conference on Neural Information Processing Systems (NeurIPS), IEEE Conference on Decision
 and Control, American Control Conference, IEEE/RSJ International Conference on Intelligent Robots and
 Systems, IEEE Multi-Conference on Systems and Control, IFAC Conferences, Canadian Biennial Symposium
 on Communications, Journal of Dynamical and Control Systems, ICML

Book Reviews:
 Springer, CRC Press (Taylor and Francis Group), Annual Reviews in Control

INVITED TALKS AND SEMINARS (excluding most conference talks)

Invited Speaker, Mathematical Foundations of AI and Data-Driven Scientific Computing,
National University of Singapore, Singapore August 2026
Semi-plenary speaker, 27th International Symposium on Mathematical Theory of Networks and
Systems, Waterloo, Canada August 2026
Mini-course, Chair of Intelligent Control Systems, RWTH Aachen University January 2026
Invited speaker, Allerton Conference, University of Illinois, Urbana-Champaign, September 2025
Seminar talk, Institute of Automatic Control, Leibniz University Hannover, July 2025
Invited workshop speaker, European Control Conference, Thessaloniki, Greece June 2025

<i>Invited speaker, Summer school at the Dutch Institute of Systems and Control</i>	
TU Delft, Netherlands,	June 2025
<i>Seminar talk, Department of Mathematics, Louisiana State University</i>	May 2025
<i>Seminar talk, Probability seminar, Institute of Stochastics</i>	
<i>Karlsruhe Institute of Technology</i>	April 2025
<i>Seminar talk, Department of Mathematics, Universita di Padova</i>	January 2025
<i>Mini-course, Chair of Intelligent Control Systems, RWTH Aachen University</i>	January 2025
<i>Invited talk, International Symposium on Mathematical Theory of Networks and Systems</i>	
Cambridge, UK,	August 2024
<i>Mini-course, Queen's Mathematics Summer School</i>	June 2024
<i>Keynote speaker, Reinforcement Learning Symposium</i>	
The Rafik B. Hariri Institute Center for Computing & Data Sciences	
Boston University	May 2024
<i>Colloquium, University of Washington St. Louis</i>	April 2024
<i>Lecture series, University of Bologna</i>	December 2023
<i>Invited talk, International Congress on Industrial and Applied Mathematics, Tokyo</i>	August 2023
<i>Invited speaker, Banff International Research Station, Banff</i>	June 2023
<i>Geometry, Topology and Control System Design</i>	
<i>Colloquium, University of California, San Diego</i>	June 2023
<i>Invited speaker, Information Theory and Applications Workshop, San Diego</i>	February 2023
<i>Seminar talk, University of Colorado, Boulder</i>	December 2022
<i>Seminar talk, University of Illinois, Urbana-Champaign</i>	October 2022
<i>Seminar talk, University of Bologna</i>	September 2022
<i>Seminar talk, École des Mines de Paris</i>	May 2022
<i>Applied Math Colloquium, University of California, Los Angeles</i>	May 2022
<i>Seminar talk, University of California, Riverside</i>	April 2022
<i>Seminar talk, University of California, Santa Barbara</i>	January 2022
<i>Seminar talk, Center for Artificial Intelligence, RWTH Aachen University</i>	November 2021
<i>Seminar talk, University of California Los Angeles</i>	October 2021
<i>Seminar talk, University of California Los Angeles</i>	November 2021
<i>IEEE-Eta Kappa Nu, Gamma Chapter</i>	
<i>Seminar talk, Technion</i>	April 2021
<i>Seminar talk, Grenoble</i>	February 2021
<i>Seminar talk, Technical University of Munich</i>	January 2021
<i>Seminar talk, Peking University</i>	August 2020
<i>Colloquium, University of Maryland</i>	October 2020
<i>Booz Allen Hamilton Distinguished Colloquium Lecture</i>	
<i>Seminar talk, University of Passau</i>	July 2020
<i>Seminar talk, ETH Zürich</i>	January 2020
<i>Colloquium, University of California Los Angeles</i>	January 2020
<i>Seminar talk, University of Stuttgart,</i>	October 2019
<i>Colloquium talk, University of Waterloo</i>	September 2019
<i>Colloquium talk, Queen's University</i>	September 2019
<i>Plenary talk, CAIMS Annual meeting, Whistler</i>	June 2019
<i>Canadian Society of Applied and Industrial Mathematics</i>	
<i>Seminar talk, University of California, San Diego</i>	May 2018
<i>Colloquium, University of Stuttgart</i>	July 2017
<i>Colloquium, University of Toronto</i>	April 2017
<i>Seminar talk, University of Salento, Lecce</i>	June 2016
<i>Colloquium, University of Stuttgart</i>	June 2016
<i>Colloquium, University of Colorado, Boulder</i>	April 2016
<i>Invited speaker, Allerton Conference Monticello, Illinois</i>	October 2016
<i>Communication, Control, and Computing</i>	
<i>Invited speaker, Allerton Conference Monticello, Illinois</i>	October 2015

<i>Communication, Control, and Computing</i>	
Seminar talk, Concordia University	December 2014
Invited talk, McMaster University	December 2014
<i>The Canadian Mathematical Society Meeting,</i>	
Invited speaker, Allerton Conference Monticello, Illinois	October 2014
<i>Communication, Control, and Computing</i>	
Invited speaker, Banff International Research Station	October 2014
<i>Optimal Cooperation, Communication, and Learning in Decentralized Systems</i>	
Invited speaker, University of Waterloo	May 2014
<i>Sixth Biannual Meeting on Systems and Control Theory</i>	
Seminar talk, McGill University	November, 2013
<i>Centre for Intelligent Machines</i>	
Invited seminar, University of Washington St. Louis	April 2013
Invited seminar, Technical University of Munich	April 2013
Invited seminar, Queen's University	March 2013
Invited seminar, Harvard University	March 2013
Invited seminar, University of California Los Angeles	March 2013
Invited seminar, Groningen University	February 2013
Invited seminar, Georgia Institute of Technology	February 2013
Invited seminar, Utah State University	February 2013

INVITED RESEARCH VISITS/WORKSHOPS AND SYMPOSIUMS

<i>IFAC World Congress, Busan, Republic of Korea</i>	August 2026
<i>European Control Conference, Reykjavík, Iceland</i>	July 2026
<i>Nonlinear Evolution Equations and Control: celebrating Alberto's 70th birthday, Padua, Italy</i>	June 2026
<i>Optimal control theory and sub-Riemannian geometry,</i>	
<i>Sorbonne Université, Paris,</i>	May 2025
<i>Research visit, UCLA</i>	March 2025
<i>Southern California Control Workshop, UCLA</i>	April 2024
<i>Southern California Applied Math Symposium, UC Irvine</i>	April 2023
<i>Information Theory and Applications Workshop, UC San Diego</i>	Jan 2023
<i>Southern California Control Workshop UC Irvine</i>	April 2022
<i>Research Visit RWTH Aachen University</i>	November 2021
<i>Learning for Dynamics and Control, MIT</i>	May 2019
<i>Princeton Day of Optimization, Princeton University</i>	Sep 2018
<i>Research Visit, Electrical and Computer Engineering, UC San Diego</i>	June 2018
<i>Visiting Faculty, Institute for Systems Theory and Automatic Control, University of Stuttgart</i>	July 2017
<i>Jean-Michel Coron's 60th birthday, Institute Henri Poincare, Paris, France</i>	June 2016
<i>Visiting Faculty, Institute for Systems Theory and Automatic Control, University of Stuttgart</i>	July 2016
<i>Research Visit, Department of Electrical, Computer, and Energy Engineering,</i>	
<i>University of Colorado, Boulder</i>	April 2016
<i>Nonlinear Control and Geometry, Banach Centre, Poland</i>	August 2015
<i>Optimal Cooperation, Communication, and Learning in Decentralized Systems,</i>	
<i>Banff International Research Station</i>	October 2014
<i>Research Visit, Mechanical and Aerospace, UC San Diego, San Diego</i>	June 2014
<i>Symposium on Mathematical Theory of Networks and Systems, Groningen</i>	July 2014
<i>The 27th Biennial Symposium on Communications, Kingston</i>	May 2014
<i>Sixth Biennial Meeting on Systems and Control Theory, University of Waterloo</i>	May 2014
<i>The 2nd Midwest Workshop on Control and Game Theory, University of Notre Dame</i>	2013
<i>Workshop on Information and Decision in Social Networks, MIT</i>	2012
<i>The 1st Midwest Workshop on Control and Game Theory, UIUC</i>	2012
<i>Symposium on Emerging Topics in Control and Modelling, UIUC</i>	2012

<i>3rd IFAC Workshop on Distributed Estimation and Control in Networked Systems</i> , Santa Barbara	2012
<i>Initiative for Mathematical Sciences and Engineering</i> , UIUC	, 2012
<i>Midwest Workshop on Control and Game Theory</i> , UIUC	2012
<i>Southern California Control Workshop</i> , UC Los Angeles	2012
<i>Southern California Control Workshop</i> , UC Riverside	2011
<i>Southern California Control Workshop</i> , UC Santa Barbara	2010
<i>Symposium on Mathematical Theory of Networks and Systems</i> , Blacksburg	2006
<i>A Celebration of Raoul Bott's Legacy in Mathematics</i> , Montreal	2007

MEMBERSHIPS

- American Mathematic Society (AMS)
- Professional Engineering Ontario (PEO)
- Institute for Operations Research and the Management Sciences (INFORMS)
- The Canadian Applied and Industrial Mathematics Society (CAIMS)
- Society of Industrial and Applied Mathematics (SIAM)
- Institute of Electrical and Electronics Engineers (IEEE)
- IEEE Control Systems Society

TEACHING ACTIVITIES

<i>Mini-course, Short Course on Reinforcement Learning (10 hours)</i> <i>RWTH, Aachen</i>	<i>January 2026</i>
<i>Math 112, Linear Algebra, Queen's</i>	<i>Fall 2025</i>
<i>Mini-course, Short Course on Reinforcement Learning (10 hours)</i> <i>RWTH, Aachen</i>	<i>January 2025</i>
<i>Math 112, Linear Algebra, Queen's</i>	<i>Fall 2024</i>
<i>Mini-course, Topics in Machine Learning (9 hours), Queen's</i>	<i>Summer 2024</i>
<i>ECE239AS, Decision Making in Stochastic Systems, UCLA</i>	<i>Spring 2024</i>
<i>ECE141, Principles of Feedback Control, UCLA</i>	<i>Winter 2024</i>
<i>ECE239AS, Linear Time-Varying Dynamical Systems, UCLA</i>	<i>Fall 2023</i>
<i>ECE242, Nonlinear Dynamics, UCLA</i>	<i>Winter 2023</i>
<i>ECE239AS, Decision Making in Stochastic Systems, UCLA</i>	<i>Fall 2022</i>
<i>ECE239AS, Control, Identification, and Learning Algorithms, UCLA</i>	<i>Spring 2022</i>
<i>ECE141, Principles of Feedback Control, UCLA</i>	<i>Winter 2022</i>
<i>MTHE 332, Introduction to Control Systems, Queen's</i>	<i>Winter 2019</i>
<i>MTHE 430, Control Systems, Queen's</i>	<i>Fall 2018</i>
<i>MTHE 332, Introduction to Control Systems, Queen's</i>	<i>Winter 2018</i>
<i>APSC 174, Linear Algebra, Queen's</i>	<i>Winter 2018</i>
<i>MATH 932, Calculus of Variations and Optimal Control, Queen's</i>	<i>Winter 2017</i>
<i>MTHE 332, Introduction to Control Systems, Queen's</i>	<i>Winter 2017</i>
<i>APSC 174, Linear Algebra, Queen's</i>	<i>Winter 2017</i>
<i>MTHE 434, Optimization Theory and Applications, Queen's</i>	<i>Fall 2016</i>
<i>APSC 200, Engineering Design Project, Queen's</i>	<i>Fall 2016</i>
<i>MTHE 332, Introduction to Control Systems, Queen's</i>	<i>Winter 2016</i>
<i>APSC 174, Linear Algebra, Queen's</i>	<i>Winter 2016</i>
<i>MTHE 434, Optimization Theory and Applications, Queen's</i>	<i>Fall 2015</i>
<i>APSC 200, Engineering Design Project, Queen's</i>	<i>Fall 2015</i>
<i>APSC 174, Linear Algebra, Queen's</i>	<i>Winter 2015</i>
<i>MTHE 430, Control Systems, Queen's</i>	<i>Fall 2014</i>

APSC 200, Engineering Design Project, <i>Queen's</i>	Fall 2014
MTHE 335, Methods in Applied Mathematics (Distribution Theory), <i>Queen's</i>	Winter 2014
APSC 200, Engineering Design Project, <i>Queen's</i>	Fall 2013
MTHE 334, Methods in Applied Mathematics (Signals and Systems), <i>Queen's</i>	Fall 2013
ECE 580, Optimization by vector space methods, <i>UIUC (Guest lecturer)</i>	Fall 2012
MATH 235, Differential Equations, <i>Queen's</i>	Fall 2008
MATH 235, Differential Equations, <i>Queen's</i>	Fall 2007

SUPERVISORY ACTIVITIES

Current Group Members:

- *Postdoctoral Researchers:*

1. **Dr. Liangbing Luo** (Queen's) 2024-
Topic: hypoelliptic operators/Sobolev inequalities/probability theory

- *Graduate Students:*

1. **Negin Rahimiyazdi**, Ph.D. (Queen's), 2026-
Topic: TBA
2. **Thea Backhouse**, M.Sc. (Queen's), 2026-
Topic: TBA
3. **Evan Arsenault**, Ph.D. (Queen's), 2025-
Topic: TBA
4. **Mitalee Oza**, Ph.D. (UCLA), 2024-
Topic: Collapse in Large Language Models
5. **Jinghan Yu**, M.Sc. (Queen's), 2024-
Topic: Stochastically Generated Threshold Graphs
6. **Amirreza Neshaei Moghaddam**, Ph.D (UCLA), 2022-
Topic: Convergence-rate analysis in reinforcement learning
7. **Michela Lapenna**, Visiting Ph.D (Bologna), 2022-
Topic: Dynamical system view of neural networks and transformers
8. **Simone Baroncini**, Visiting Ph.D (Bologna), 2025-
Topic: Geometric studies reinforcement learning

Alumni:

- *Postdoctoral Researchers:*

1. **Mohammad Akbari**, (online learning and control) 2022-2023
Current Placement: Machine learning consultant at IDIN Consulting
2. **Kexue Zhang**, (delayed differential equations, event-triggered control systems) 2017-2019
Current Placement: Assistant Professor, jointly with Department of Mathematics and Statistics, and Smith School of Engineering, Queen's University
3. **Robert Kipka**, (co-advised, infinite-dimensional optimal control), 2014-2016
Current Placement: Associate Professor, Department of Mathematical Sciences, Lake Superior State University

- *Graduate Students:*

1. **Xiaoqing Li**, M.Sc. (RTWH Aachen), 2025-2026
Topic: Robustness properties of flexible-step MPC

2. **Ala Kolsi**, M.Sc. (RTWH Aachen), 2025-2026
Topic: Flexible-step MPC for nonholonomic systems
3. **Henri Cimetiere**, Visiting M.Sc. (Ecol des Pont), Winter 2025
Topic: Clustering models in machine learning
4. **Arijit Dey**, M.Sc. independent project (UCLA), 2025
Topic: Convergence of asynchronous stochastic gradient descent under delay
5. **Markus Pietschner**, M.Sc. (RWTH Aachen), 2014-2025
Topic: Flexible-step Model Predictive Control for unknown systems
6. **Annika Fürnsinn**, Ph.D. 2019-2024
Topic: Generalized Lyapunov functions and model predictive control
Placement: Lecturer (permanent), TH Wildau, Berlin
7. **Somya Singh**, Ph.D. 2018-2023
Topic: Interacting Polya urns
Placement: Postdoctoral scholar, UC Louvain $\Rightarrow$ IIT Bombay
 $\Rightarrow$ Assistant Professor (tenure track), Department of Mathematics, Shiv Nadar University
8. **Justin Veiner**, M.Sc. 2021-2023
Topic: Generative adversarial networks based on a general parameterized family of loss functions
9. **Daniel Adu Owusu**, Ph.D. 2017-2022
Topic: Optimal transport in mechanism design, differential games and ensemble control
Placement: Assistant Professor (limited term), Department of Mathematics, University of Georgia
 $\Rightarrow$ Assistant Professor (tenure track), Department of Mathematics and Statistics, Villanova University
10. **Mohammad Akbari**, Ph.D., 2017-2022
Topic: Online learning in control theory
Placement: Postdoctoral scholar, Queen's University
 $\Rightarrow$ Machine learning consultant at IDIN Consulting
11. **Adam Gronowski**, M.Sc. 2020-2022
Topic: Information bottleneck methods for fairness and privacy in machine learning
Awards: Vector Institute Scholarship
Canadian Society of Information Theory Best Paper Award *Placement*: co-founder of caps AI
12. **Xu Chen**, M.Sc. (with C. Ebenbauer at Stuttgart University), 2020-2021
Thesis: Stability analysis of the ADAM algorithm
Placement: Ph.D. student, *Aachen University*
13. **Himesh Bhatia**, M.Sc. 2018-2020
Topic: Generative adversarial networks
Awards: Dorrance Family Award
14. **Greg Harrington**, M.Sc. 2017-2020
Topic: Optimization policies for Polya contagion networks)
Awards: Queen's Frank E Smith, Queen Elizabeth II Graduate Scholarship in Science and Technology *Placement*: Ciena
15. **Michael McCreesh**, M.Sc., 2017-2019
Topic: Accelerated convergence of saddle-point dynamics
Awards: NSERC, Queen's Tri Council, OGS
Placement: Ph.D. student, *Mechanical and Aerospace Engineering, UC San Diego*
16. **Simon Michalowsky**, visiting Ph.D. student 2015-2019
Institute for Systems Theory and Automatic Control (University of Stuttgart)
Topic: Lie bracket approximation for distributed optimization

17. **Scott Kyle**, M.Sc. 2017-2019
 Topic: Control of nonholonomic mechanical systems using virtual surfaces
 Placement: MacLean Engineering
18. **Diego Cardenas**, M.Sc. 2018-2019
 Topic: On the Kalman filter with intermittent observations
 Placement: Ph.D. student, *University of Agder, Norway*
19. **Connor Boyd**, M.Sc. 2017-2018
 Topic: Vakonomic and non-holonomic dynamics Placement: Machine Learning Developer, Blue-Cat
20. **Alireza Samsamshariat**, M.Sc. 2017-2018
 Topic: Martingale optimal transportation
 Placement: Data Scientist, Zynga
21. **Mikhail Hayhoe**, M.Sc. 2015-2017
 Thesis: A Polya urn stochastic model for the analysis and control of epidemics on networks
 Awards: **ACC Best Paper Award Finalist**, Queen's F. E. Smith, NSERC Queen's Graduate Award
 Placement: Ph.D. student, *Electrical and Systems Engineering, University of Pennsylvania*
 ⇒ Principal Data Scientist at Amgen
22. **Jeremy Coulson**, M.Sc. 2015-2017
 Thesis: Average controllability of random heat equations
 Awards: NSERC, Queen's Tri Council, OGS, Queen's H. K. Walter
 Placement: Ph.D. student, *Automatic Control Laboratory, ETH Zurich*
 ⇒ Assistant Professor, Electrical & Computer Engineering, University of Wisconsin-Madison
23. **Drew Steeves**, M.Sc. 2015-
 Thesis: Controllability of underactuated coupled parabolic systems
 Awards: Queen's Senator Frank Carrel, Queen's Graduate Award, Queen Elizabeth II Graduate Scholarship in Science and Technology
 Placement: Ph.D. student, *Mechanical and Aerospace Engineering, UC San Diego*
 ⇒ AV Planning & Prediction, Nissan
24. **Pouya Rezaeinia**, M.Sc. 2015-2017
 Thesis: Push-sum algorithm on time-varying random graphs
 Placement: Ph.D. student, *Sauder Business School, UBC*
 ⇒ Management Consultant at CPCS
25. **Daniel Adu Owusu**, M.Sc., 2013-2014
 Project: *On ensemble control systems*
 Placement: Ph.D. student *Mathematics, Queen's University* Limited term Assistant Professor, Department of Mathematics, University of Georgia
26. **Mohammad Akbari**, M.Sc. 2013-2015
 Thesis: Distributed online optimization on time-varying networks
 Placement: Ph.D. student *Mathematics, Queen's University*
27. **Babak Beheshti Vadeqan**, M.Sc. 2014-2016
 Thesis: Geometry of Dirac Operators,
 Placement: Ph.D. student *Mathematics, Western University*
28. **Ali Khanafer**, Ph.D. (visiting student from UIUC), 2012-2014
 Thesis: Information spread in networks: games, optimal control, and stabilization
 Placement: *Staff Machine Learning Engineer, Shopify*

- Undergraduate Research Students:

- *Undergraduate Research:*
 - ◇ **2015-2026:**
 - Elizabeth Bighiu (Undergraduate Research Assistant)
Project: Reinforcement learning for recommender systems
 - ◇ **2019-2020:**
 - Rebecca Bonham Carter (NSERC USRA)
 - Fernando Camacho (NSERC USRA)
Project: Rough path theory
 - ◇ **2018-2019:**
 - Ian Hogeboom-Burr (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - Bryony Schonewille (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - Hugh Corley (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - ◇ **2017-2018:**
 - Himesh Bhatia (NSERC USRA)
Project: Passivity and self-appraisal dynamics
 - Daniel Tamming (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - Fernando Camacho (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - ◇ **2016-2017:**
 - Rebecca Bonham-Carter (NSERC USRA)
Project: Extensions of sparse stable matrix systems
 - Michael McCreesh (NSERC USRA)
Project: Metzler sparse stable matrix cones
 - Himesh Bhatia (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - Stephen Chisnall (Summer Work Experience Program)
Project: Experimental design of leader-follower robotics laboratory
 - ◇ **2015-2016:**
 - Drew Steeves (Summer Work Study Program)
Project: Re-design of the Modern Control Laboratory
 - ◇ **2014-2015:**
 - Drew Steeves (Summer Work Experience Program)
Project: Convergence rate of one-sided asymmetric Hegselmann-Krause dynamics
 - Jeremy Coulson (Summer Work Experience Program)
Project: Convergence rate of one-sided asymmetric Hegselmann-Krause dynamics
- *Undergraduate Thesis Students:*
 - ◇ **2018-2019:**
 - Lachlan Devir, Chris Caromicoli, Andrew Downie, Alex Taylor (*Keyser Prize for best project*)
Thesis: Motion planning in dynamic environments
 - Luke Chau, Simon Kersten, Thomas Huckell, Benji Christie *Thesis: Predictive control for an autonomous boat*

- Coby Davis, Ted Bursey, Julian Gailiunas, Jason Benchetrit *Thesis: Predictive control for transportation*
- Adam Boljkovac, Steven Lee, Patrick Chin, Jordana Sherman Motion planning in dynamic environments
- ◇ **2016-2017:**
 - Cleo Savides, Joshua Dirocco, Mareena Macpherson (*Keyser Prize for best project*)
Thesis: Underwater source seeking using a deformable fish
 - Greg Harrington, Sarah Colquhoun, Kerem Ataman, Tristen Ognibene
Thesis: Analysis and control of contagious phenomena in networks
- ◇ **2015-2016:**
 - Taylor Reynolds, True Wilson, Matthew Boyd (*Keyser Prize for best project*)
Thesis: Extremum seeking control methods for nonholonomic source seeking vehicles
 - Jacob Malleau, Soraya Weaver, Rehman Shivji, Aaron Short
Thesis: Dynamic vehicle routing for UBER
- ◇ **2014-2015:**
 - Drew Steeves, Jeremy Coulson, Ted Donnelly, Thomas Hall (*Keyser Prize for best project*)
Thesis: Synchronization of coupled oscillators: power networks
 - Ryan Farrell, Mikhail Hayhoe, Justin Ma, Mark Mahony
Thesis: Distributed triggering strategies for deployment of autonomous mobile networks with outdated information
 - Matthew Slavin, Michael Mehta, Kendra MacKay, Eve Lavery (co-supervised)
Thesis: Formation dynamics
 - John Ramsey, Daniel Potvin, Brendon Conlin (co-supervised)
Thesis: Extending a Polya contagion process to modelling contagion in networks with finite memory
 - Hannah Koke, Brynn Vadala (co-supervised)
Thesis: Profiling Contagious Behaviour in Networks
- ◇ **2013-2014:**
 - Marlee Vandewouw, Karin Martin (*Keyser Prize for best project*)
Thesis: Distributed algorithms for deployment of autonomous mobile networks
 - Ian Ross, Evangelian Collings, Taylor Adams
Thesis: Formation dynamics of multi-agent systems

THESIS COMMITTEE/CANDIDACY EXAM MEMBER

- Antoine Legat, Ph.D., *ICTEAM, UCLouvain*, 2025
- Matteo Marchi, Ph.D., *UCLA*, 2025
- Zida Wu, Ph.D., *UCLA*, 2025
- David Thorne, Ph.D., *UCLA*, 2025
- Hazhir Aliahmadi, Ph.D., *Queen's University*, 2025
- Jasper Verbree, Ph.D., *University of Groningen*, 2024
- Mona Buisson Fenet, Ph.D., *École des Mines de Paris*, 2023
- Aaron John Sabu, M.Sc., *UCLA*, 2023
- Kangdi Yu, M.Sc., *UCLA*, 2023
- Jonathan Burton, Ph.D., *UCLA*, 2022
- Lucas Fraile, Ph.D., *UCLA*, 2022

- Derek Xioa, Ph.D., *UCLA*, 2022
- Muratkhan Abdirash, Ph.D., *UCLA*, 2022
- Zida Wu, Ph.D., *UCLA*, 2022
- Jan Feiling Ph.D., *University of Stuttgart*, 2021
- Ahmed Shaltut, M.Sc., *Queen's University*, 2021
- Xin Cheng, Ph.D., *Queen's University*, 2021
- Mohammad Yaali Jahromi, M.Sc., *Queen's University*, 2020
- Bora Yongacoglu, Ph.D., *Queen's University*, 2018
- Sina Sanjari, Ph.D., *Queen's University*, 2018
- Yanlei Zhang, Ph.D., *Queen's University*, 2018-
- Graeme Baker, M.Sc., *Queen's University*, 2018
- Sina Sanjari, M.Sc., *Queen's University*, 2017
- Kiraseya Preusser, M.Sc., *Queen's University*, 2017
- Iman Askarian, M.Sc., *Queen's University*, 2016
- Tristan Mines, M.Sc., *Queen's University*, 2016
- Francois Seguin, Ph.D., *Queen's University*, 2016
- Judith Ebegbulem, M.Sc., (Chem-Eng), *Queen's University*, 2016
- Ryan Bennett, M.Sc. (Chem-Eng), *Queen's University*, 2015
- Khaled Hayajneh, Ph.D. (ECE), *Queen's University*, 2015
- Arghavan Modiri, M.Sc. (ECE), *Queen's University*, 2015
- Michael Cabral, M.Sc., *Queen's University*, 2014
- Hossein Mousavian, Ph.D., *Queen's University*, 2014
- Sousan Beigi Harchegani, M.Sc. (ECE), *Queen's University*, 2014
- Naci Saldi, Ph.D., *Queen's University*, 2013
- Saber Jafarpour, Ph.D., *Queen's University*, 2013
- Emmanuel Ogbe (Chem-Eng), Ph.D., *Queen's University*, 2013
- Joshua Pohlkamp-Hartt, Ph.D., *Queen's University*, 2013
- Youness Aliyari, Ph.D., *Queen's University*, 2013
- Melkior Ornik, M.Sc., *Queen's University*, 2013

JOURNAL PUBLICATIONS

1. Alvaro Rodriguez Abella, Joao Silvestre, *Bahman Ghahsifard*, and Paulo Tabuada, Deep learning from a control vantage point, **Control Systems Magazine**, to appear, 2026
2. *Bahman Ghahsifard*, On the topology of neural network superlevel sets, **Comptes Rendus, Série Mathématique**, submitted, 2026
3. Shamak Dutta, *Bahman Ghahsifard*, and Stephen Smith, Submodular optimization with applications to decision and control, **Automatica**, submitted, 2026
4. Daniel Adu, Karthik Elamvazhuti, *Bahman Ghahsifard*, From Schrödinger bridge to optimal transport over sub-Riemannian manifolds **SIAM Journal on Control and Optimization**, submitted, 2026

5. Ala Kolsi, Christian Ebenbauer, *Bahman Ghahesifard*, Raik Suttner, Global exponential stabilization of a force- and torque-actuated unicycle by flexible-step MPC, **Automatica**, submitted, 2026
6. Michela Lapenna, Rita Fioresi, and *Bahman Ghahesifard*, Sinkhorn doubly stochastic attention rank decay analysis, **Transactions on Machine Learning Research**, conditionally accepted, 2026
7. Arijit Dey and *Bahman Ghahesifard*, Graph State Networks (GSNs): persistent node-wise selective State Space Models, **Transactions on Machine Learning Research**, 2026
8. Simone Baroncini, *Bahman Ghahesifard*, and Giuseppe Notarstefano, On reward-balancing in Reinforcement Learning, **IEEE Transaction on Automatic Control**, submitted, 2026
9. Evan Arsenault, Amirreza Neshaei Moghaddam, and *Bahman Ghahesifard*, Geometric homogeneity in temporal difference learning, **IEEE Control Systems Letters**, to appear, 2026
10. Jinghan Yu, Fady Alajaji, and *Bahman Ghahesifard*, Pólya Threshold Graph, **Network Science**, Cambridge University Press, submitted, 2026
11. Amirreza Neshaei, Alex Olshevsky, and *Bahman Ghahesifard*, Sample complexity of the linear quadratic regulator: a reinforcement learning lens, **Journal of Machine Learning Research**, 26, 1–50, 2025
12. Henri Cimetiere, Maria Teresa Chiri, and *Bahman Ghahesifard*, Localmax dynamics for attention in transformers and its asymptotic behavior, **SIAM Journal on Control and Optimization**, submitted, 2025
13. Amirreza Neshaei, Alex Olshevsky, and *Bahman Ghahesifard*, Sample complexity of linear quadratic regulator without initial stability, **IEEE Transaction on Automatic Control**, submitted, 2025
14. Amirreza Neshaei and *Bahman Ghahesifard*, Gradient flow approximations in temporal difference learning, **IEEE Control Systems Letters**, 8, 2655–2660, 2024
15. Pouya Rezaenia and *Bahman Ghahesifard*, Tamas Linder, Distributed optimization with uncertain communications, **IEEE Transactions on Automatic Control**, 70 (4), 2746–2753, 2025
16. *Bahman Ghahesifard* and Christian Ebenbauer, An isospectral saddle-point dynamics, **AIMS Mathematics**, special issue dedicated to Roger Brockett, to appear, 2025
17. Paulo Tabuada and *Bahman Ghahesifard*, Corrigendum to “Universal Approximation Power of Deep Residual Neural Networks Through the Lens of Control”, **IEEE Transactions on Automatic Control**, PP(99):1–2, 2024
18. Annika Fürnsinn, Christian Ebenbauer, and *Bahman Ghahesifard*, Flexible-step MPC for switched linear systems with no quadratic common Lyapunov function, **IEEE Transactions on Automatic Control**, to appear, 2025
19. Xudong Chen and *Bahman Ghahesifard*, Sparse linear ensemble systems and structural averaged controllability: single-input case, **IEEE Transactions on Control of Network Systems**, 12(3), 2314–2324, 2025
20. Daniel Adu and *Bahman Ghahesifard*, Approximate controllability of continuity equation of transformers, **IEEE Control Systems Letters**, 8, 964–969, 2024
21. Annika Fürnsinn, Christian Ebenbauer, and *Bahman Ghahesifard*, Flexible-step model predictive control based on generalized Lyapunov functions, **Automatica**, 175, 112215, 2025

22. Alex Olshevsky and *Bahman Ghahsifard*, A small-gain analysis of single timescale actor critic, **SIAM Journal on Control and Optimization**, 61, 2710–2738, 2023
23. Mohammad Akbari, Amirreza Neshaei, *Bahman Ghahsifard*, and Tamas Linder, Logarithmic regret for adaptive control of linear quadratic regulators using hints, **IEEE Transactions on Automatic Control**, revised, 2025
24. Justin Veiner, Fady Alajaji, and *Bahman Ghahsifard*, A unifying generator loss function for generative adversarial networks, **Entropy**, 26 (4): 290, 2024
25. Somya Singh, Fady Alajaji, and *Bahman Ghahsifard* A preferential attachment model based on a Pólya urn with increasing colors, **Network Science**, Cambridge University Press, 1-21, 2024,
26. Annika Fürnsinn, Christian Ebenbauer, and *Bahman Ghahsifard*, Relaxed feasibility and stability criteria for flexible-step MPC, **IEEE Control Systems Letters**, 7, 2851–2856, 2023
27. Daniel Adu and *Bahman Ghahsifard*, Robust matching for teams, **Journal of Optimization Theory and its Applications**, 501-523, 2024
28. Matteo Marchi, Jonathan Bunton, *Bahman Ghahsifard*, Paulo Tabuada, Sharp Performance Bounds for PASTA, **IEEE Control Systems Letters**, 7, 2401–2406, 2023
29. Behrouz Touri and *Bahman Ghahsifard*, A united framework for continuous-time distributed optimization, **SIAM Journal on Control and Optimization**, 62(1): 1–27, 2024
30. Adam Gronowski, William Paul, Fady Alajaji, *Bahman Ghahsifard*, and Philippe Burlina, Classification Utility, Fairness, and Compactness via Tunable Information Bottleneck and Rényi Measures, **IEEE Transactions on Information Forensics and Security**, 19: 5730–5745, 2024
31. Paulo Tabuada and *Bahman Ghahsifard*, Universal approximation power of deep residual neural networks through the lens of control, **IEEE Transactions on Automatic Control**, 68(10): 6137–6152, 2023
32. Kexue Zhang, *Bahman Ghahsifard*, and Elena Braverman, A note on the stability of event-triggered control systems with delay, **IEEE Transactions on Automatic Control**, 69(7): 4572–4579, 2024
33. Mohammad Akbari, *Bahman Ghahsifard*, and Tamas Linder, Logarithmic regret in online linear quadratic control using Riccati updates, **Mathematics of Control, Signals, and Systems**, 34: 59–90, 2022
34. Karthik Elamvazhuti, *Bahman Ghahsifard*, Andrea Bertozzi, and Stanley Osher, Neural ODE control for trajectory approximation of continuity equation, **IEEE Control Systems Letters**, 3152 – 3157, 2022
35. Matteo Marchi, Johnathan Bunton, *Bahman Ghahsifard*, and Paulo Tabuada, Stability guarantees for control loops with deep learning state estimation, **IEEE Control Systems Letters**, 6, 1286 – 1291, 2021
36. Daniel Adu, Tamer Başar, and *Bahman Ghahsifard*, Optimal transport for a class of linear quadratic differential games, **IEEE Transactions on Automatic Control**, 67(11), 6287 - 6294, 2022
37. Andrew Downie, *Bahman Ghahsifard*, and Stephen Smith, Submodular maximization with limited function access, **IEEE Transactions on Automatic Control**, PP(99): 1–14, 2023

38. Somya Singh, Fady Alajaji, and *Bahman Gharesifard* Consensus Using a network of finite memory Polya Urns, **IEEE Control Systems Letters**, 2780 – 2785, 2022
39. *Bahman Gharesifard*, and Xudong Chen, Structural averaged controllability of linear ensemble systems, **IEEE Control Systems Letters**, 6, 518 – 523, 2021
40. Kexue Zhang and *Bahman Gharesifard*, and Elena Braverman, Event-triggered control for discrete-time delay systems, **Automatica**, 147, 110688, 2023
41. Mohammad Akbari, *Bahman Gharesifard*, and Tamas Linder, On the lack of monotonicity of Newton-Hewer updates for Riccati equations, **Automatica**, 132, 109788, 2021
42. Somya Singh, Fady Alajaji, and *Bahman Gharesifard*, A finite memory interacting Polya contagion network and its approximating dynamical systems, **SIAM Journal on Control and Optimization**, 60(2), 347–369, 2022
43. Greg Harrington, Fady Alajaji, and *Bahman Gharesifard*, Initialization and curing policies for Polya contagion networks, **SIAM Journal on Control and Optimization**, 60(2), 170–195, 2022
44. Meriem Gharbi, *Bahman Gharesifard*, and Christian Ebenbauer Anytime proximity moving horizon estimation: stability and regret, **IEEE Transactions on Automatic Control**, 68 (6), 2023
45. Kexue Zhang and *Bahman Gharesifard*, and Elena Braverman, Event-triggered control for nonlinear time-delay systems, **IEEE Transactions on Automatic Control**, 67(2): 1031-1037, 2022
46. Kexue Zhang and *Bahman Gharesifard*, Hybrid event-triggered and impulsive control for time-delay systems, **Nonlinear Analysis: Hybrid Systems**, 43: 101109, 2022
47. Himesh Bhatia, William Paul, Bahman Gharesifard, Fady Alajaji, and Philippe Burlina, Renyi Generative Adversarial Networks, **Neural Computation**, 1-38, 2021
48. Xudong Chen and *Bahman Gharesifard*, Distinguished sets for semi-simple Lie algebras, **Journal of Algebraic Combinatorics**, 1-13, 2021
49. Behrouz Touri and *Bahman Gharesifard*, A modified saddle-point dynamics for distributed convex optimization on general directed graphs, **IEEE Transactions on Automatic Control**, 65-7, 3098-3103, 2020
50. Simon Michalowsky, *Bahman Gharesifard*, and Christian Ebenbauer, A Lie bracket approximation approach to distributed optimization, **Automatica**, 112, 108691, 2020
51. Mikhail Hayhoe, Fady Alajaji, and *Bahman Gharesifard*, Curing epidemics on networks using a Polya contagion model, **IEEE Transactions on Networking**, 27(5), 2085-2097, 2019
52. Mohammad Akbari, *Bahman Gharesifard*, and Tamas Linder, Individual regret bounds for the distributed online alternating direction method of multipliers, **IEEE Transactions on Automatic Control**, 64(4), 1746-1752, 2019
53. Pouya Rezaenia, *Bahman Gharesifard*, Tamas Linder, and Behrouz Touri, Push-sum on random graphs: almost sure convergence and convergence rate, **IEEE Transactions on Automatic Control**, 65(3), 1295-1302, 2019

54. Drew Steeves, *Bahman Ghahesifard* and Abdol-Reza Mansouri, Controllability of underactuated coupled parabolic systems with many underactuators, Part II: null controllability, **SIAM Journal on Control and Optimization**, 57(5), 3272–3296, 2019 (**SIAG/CST Best SICON Paper Prize 2021**)
55. Drew Steeves, *Bahman Ghahesifard* and Abdol-Reza Mansouri, Controllability of underactuated coupled parabolic systems with many underactuators, Part I: algebraic solvability, **SIAM Journal on Control and Optimization**, 57:5, 3297–3321, 2019 (**SIAG/CST Best SICON Paper Prize 2021**)
56. Jeremy Coulson, *Bahman Ghahesifard* and Abdol-Reza Mansouri, Average controllability of random heat equations with general distributions, **Automatica**, 103, 46–52, 2019
57. Michael McCreesh and *Bahman Ghahesifard*, Stability of bounded subsets of Metzler sparse matrix cones, **Automatica**, 95, 544–547, 2018
58. Mikhail Hayhoe, Fady Alajaji, and *Bahman Ghahesifard*, A Polya contagion model for networks, **IEEE Transactions on Control of Network Systems**, 5(4), 1998–2010, 2018
59. Shreyas Sundaram and *Bahman Ghahesifard*, Distributed optimization under adversarial nodes, **IEEE Transactions on Automatic Control**, 64(3), 1063–1076, 2019
60. Aris Tsopelakos, Mohamed-Ali Belabbas, and *Bahman Ghahesifard*, Classification of the structurally controllable zero-patterns for driftless bilinear control systems, **IEEE Transactions on Control of Network Systems**, 6(1), 429–439, 2019
61. *Bahman Ghahesifard* and Stephen L. Smith, Distributed submodular maximization with limited information, **IEEE Transactions on Control of Network Systems**, 5(4), 1635–1645, 2018
62. Ashish Cherukuri, *Bahman Ghahesifard*, and Jorge Cortés, Saddle-point dynamics: conditions for asymptotic stability of saddle points, **SIAM Journal on Control and Optimization**, 55 (1), 486–511, 2017
63. Cesar Aguilar and *Bahman Ghahesifard*, Almost equitable partitions and new necessary conditions for network controllability, **Automatica**, 80, 25–31, 2017
64. *Bahman Ghahesifard*, Stabilization of bilinear sparse matrix control systems using periodic inputs, **Automatica**, 77, 239–245, 2017
65. Mohammad Akbari, *Bahman Ghahesifard*, and Tamas Linder, Distributed online convex optimization on time-varying directed graphs, **IEEE Transactions on Control of Network Systems**, Volume 4(3), 417–428, 2017
66. *Bahman Ghahesifard*, Behrouz Touri, Tamer Başar, Jeff Shamma, On the convergence of piecewise linear strategic interaction dynamics on networks, **IEEE Transactions on Automatic Control**, 61 (6), 1682–1687, 2016
67. Ali Khanafer, Tamer Başar, *Bahman Ghahesifard*, Stability of epidemic models over directed graphs: a positive systems approach, **Automatica**, 74, 126–134, 2016
68. *Bahman Ghahesifard*, Tamer Başar, and Alejandro Domínguez-García, Price-based coordinated aggregation of networked distributed energy resources, **IEEE Transactions on Automatic Control**, 61(10), 2936–2946, 2016

69. Cesar Aguilar and *Bahman Ghahesifard*, Graph controllability classes for the Laplacian leader-follower dynamics, **IEEE Transactions on Automatic Control**, 60 (6), 1-13, 2015
70. Cesar Aguilar and *Bahman Ghahesifard*, Laplacian Controllability Classes for Threshold Graphs, **Linear Algebra and its Applications**, 471, 575-586, 2015
71. *Bahman Ghahesifard* and Jorge Cortés, Continuous-time distributed optimization on directed networks, **IEEE Transaction on Automatic Control**, 59 (3), 781-786, 2014
72. *Bahman Ghahesifard* and Jorge Cortés, Stealthy deception in hypergames under informational asymmetry, **IEEE Transactions on Systems, Man, and Cybernetics, Part A**, 44 (6), 785-795, 2014
73. *Bahman Ghahesifard* and Jorge Cortés, Distributed convergence to Nash equilibria in two-network zero-sum games, **Automatica**, 49 (6), 1683-1692, 2013
74. *Bahman Ghahesifard* and Jorge Cortés, Distributed strategies for generating weight-balanced and doubly stochastic digraphs, **European Journal of Control**, 18 (6), 539-557, 2012
75. *Bahman Ghahesifard* and Jorge Cortés, Evolution of players' misperceptions in hypergames under perfect observations, **IEEE Transaction on Automatic Control**, 57 (7), 1641-1656, 2012
76. *Bahman Ghahesifard*, Stabilization of systems with one degree of underactuation with energy shaping, a geometric approach, **SIAM Journal on Control and Optimization**, 49 (4), 1422-1434, 2011
77. *Bahman Ghahesifard* and Andrew D. Lewis and Abdol-Reza Mansouri, A geometric framework for stabilization by energy shaping: sufficient conditions for existence of solutions, **Brockett Legacy Special Issue of Journal of Communications in Information and Systems**, 8(4), 353-398, 2008

CONFERENCE PROCEEDINGS

1. Markus Pietschner, Raik Suttner, *Bahman Ghahesifard*, Christian Ebenbauer, Flexible Model Predictive Control of unknown systems, Proceedings of the European Control Conference, Reykjavík, Iceland, 2026, to appear
2. *Bahman Ghahesifard*, Matteo Marchi, Jonathan Bunton, Paulo Tabuada, Preventing model collapse: A Fisher-Rao perspective on the dynamics of training with synthetic data, Proceedings of the IEEE Conference on Decision and Control, Honolulu, submitted, 2026
3. Amirreza Neshaei, Xudong Chen, and *Bahman Ghahesifard*, On structural averaged controllability of multi-input linear ensemble systems: The acyclic case, Proceedings of the IEEE Conference on Decision and Control, Honolulu, submitted, 2026
4. *Bahman Ghahesifard* and Paulo Tabuada, Preventing model collapse when training LLMs with synthetic data, Proceedings of the IEEE Conference on Decision and Control, Rio de Janeiro, Brazil, 2025 (invited paper)
5. Amirreza Neshaei and *Bahman Ghahesifard*, Gradient flow approximations in temporal difference learning, Proceedings of the American Control Conference, Denver, Colorado, 2025

6. Luc McCutcheon, *Bahman Ghahesifard*, and Saber Fallah, Neural Lyapunov function approximation with self-supervised reinforcement learning, Proceedings of the International Conference on Robotics and Automation (ICRA), 2025
7. Marc Weber, Christian Ebenbauer, *Bahman Ghahesifard*, Inferring global exponential stability properties using lie-bracket approximations, Proceedings of the IEEE Conference on Decision and Control, 2024
8. Simone Baroncini, Guido Carnevale, *Bahman Ghahesifard*, Giuseppe Notarstefano, A system-theoretic note on model-based actor-critic, Proceedings of the IEEE Conference on Decision and Control, 2024 (**The 2024 CSS TC on Robust and Complex Systems Outstanding Student Paper Prize**)
9. Daniel O Adu, *Bahman Ghahesifard*, Approximate controllability of continuity equation of transformers, Proceedings of the IEEE Conference on Decision and Control, 2024
10. Mohammad Akbari, *Bahman Ghahesifard*, Tamas Linder, Achieving logarithmic regret via hints in online learning of noisy LQR systems, IEEE Conference on Decision and Control, 2022 (**invited paper**)
11. Andrew Downie, *Bahman Ghahesifard*, Stephen Smith, A programming approach for worst-case studies in distributed submodular maximization, IEEE Conference on Decision and Control, 2023 (**invited paper**)
12. Andrew Downie, *Bahman Ghahesifard*, Stephen Smith, Optimistic greedy strategies for partially known submodular functions, IEEE Conference on Decision and Control, 2022 (**invited paper**)
13. Matteo Marchi, Jonathan Bunton, *Bahman Ghahesifard*, Paulo Tabuada, LiDAR point cloud registration with formal guarantees, IEEE Conference on Decision and Control, 2022
14. Somya Singh, Fady Alajaji, and *Bahman Ghahesifard*, Modeling network contagion via interacting finite memory Polya urns, IEEE International Symposium on Information Theory (ISIT), 2022
15. Adam Gronowski, William Paul, Fady Alajaji, *Bahman Ghahesifard*, and Philippe Burlina Rényi fair information bottleneck for image classification, 17th Canadian Workshop on Information Theory, **Canadian Society of Information Theory Best Paper Award**, 2022
16. *Bahman Ghahesifard*, and Xudong Chen, Structural averaged controllability of linear ensemble systems, Proceedings of the IEEE Conference on Decision and Control, Austin, 2021 (**invited paper**)
17. Matteo Marchi, Johnathan Bunton, *Bahman Ghahesifard*, and Paulo Tabuada, Stability guarantees for control loops with deep learning state estimation, Proceedings of the IEEE Conference on Decision and Control, Austin, 2021
18. Marc Weber, *Bahman Ghahesifard*, and Christian Ebenbauer, A note on Nussbaum-type control and Lie-bracket approximation, Proceedings of the IEEE Conference on Decision and Control, Austin, 2021
19. Meriem Gharbi, *Bahman Ghahesifard*, and Christian Ebenbauer, Anytime proximity moving horizon estimation: stability and regret, Proceedings of the IEEE Conference on Decision and Control, Austin, 2021 (**invited paper**)

20. Paulo Tabuada and *Bahman Ghahsifard*, Universal approximation power of deep residual neural networks via nonlinear control theory, **International Conference on Learning Representations (ICLR)**, 2021
21. Matteo Marchi, *Bahman Ghahsifard*, Paulo Tabuada, Training deep residual networks for uniform approximation guarantees, Learning for Dynamics and Control (L4DC), ETH Zurich, June 2021
22. Mohammad Akbari, *Bahman Ghahsifard*, Tamas Linder, Riccati updates for online linear quadratic control, Learning for Dynamics and Control (L4DC), UC Berkeley, June 2020
23. Greg Harrington, Fady Alajaji and *Bahman Ghahsifard*, Infection-curing games over Polya contagion networks, Proceedings of the Sixteenth Canadian Workshop on Information Theory, Hamilton, Canada, 2019
24. Pouya Rezaenia, *Bahman Ghahsifard*, Tamas Linder, and Behrouz Touri, Distributed optimization on random graphs, Proceedings of the 7th IFAC Workshop on Distributed Estimation and Control in Networked Systems, Groningen, Netherlands, 391-395, 2018
25. Pouya Rezaenia, *Bahman Ghahsifard*, Tamas Linder, and Behrouz Touri, Convergence rate of push-sum algorithms on random graphs, Proceedings of the IEEE Conference on Decision and Control, Miami, USA, 2018 (**invited paper**)
26. Drew Steeves, *Bahman Ghahsifard* and Abdol-Reza Mansouri, Controllability of coupled parabolic systems with multiple underactuators, Proceedings of the IEEE Conference on Decision and Control, Miami, USA, 2019
27. Simon Michalowsky, *Bahman Ghahsifard*, and Christian Ebenbauer, On the Lie bracket approximation approach to distributed optimization: Extensions and limitations, Proceedings of the European Control Conference, Cyprus, Greece, 119-124, 2018
28. Mikhail Hayhoe, Fady Alajaji, and *Bahman Ghahsifard*, Curing with the Network Polya Contagion Model, Proceedings of the American Control Conference, Milwaukee, 2644-2650, 2018
29. Xudong Chen and *Bahman Ghahsifard*, Distinguished vector fields over smooth manifolds with applications to ensemble control, Proceedings of the IEEE Conference on Decision and Control, Melbourne, Australia, 1963-1968, 2017
30. Simon Michalowsky, *Bahman Ghahsifard*, and Christian Ebenbauer, Distributed extremum seeking over directed graphs, Proceedings of the IEEE Conference on Decision and Control, Melbourne, Australia, 2095-2101, 2018 (**invited paper**)
31. Christian Ebenbauer, Simon Michalowsky, Victoria Grushkovskaya, and *Bahman Ghahsifard*, Distributed optimization over directed graphs with the help of Lie brackets, Proceedings of the 20th IFAC World Congress, Toulouse, France, 50 (1), 15343-15348, 2017
32. *Bahman Ghahsifard* and Christian Ebenbauer, Convergence properties of an Isospectral saddle-point dynamics, Proceedings of the 20th IFAC World Congress, Toulouse, France, 50 (1), 8460-8464, 2017,
33. Mikhail Hayhoe, Fady Alajaji, and *Bahman Ghahsifard*, A Polya urn-based model for epidemics on networks, Proceedings of the American Control Conference, Seattle, 358-363, 2017 (**Finalist for Best Paper Award**)

34. Behrouz Touri and *Bahman Ghahesifard*, Saddle-point dynamics for distributed convex optimization on general directed graphs, Proceedings of the IEEE Conference on Decision and Control, Las Vegas, NV, 862-866, 2016
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36. Mohammad Akbari, *Bahman Ghahesifard*, and Tamas Linder, Regret bounds for the distributed online alternating direction method of multipliers, Annual Allerton Conference on Communication, Control, and Computing, Monticello, 2016, (**invited talk**)
37. Mohamed-Ali Belabbas and *Bahman Ghahesifard*, On the structural controllability of sparse bilinear control systems, Proceedings of the 22nd International Symposium on Mathematical Theory of Networks and Systems, Minneapolis, Minnesota, 2016
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40. *Bahman Ghahesifard*, Averaging methods for stabilization of bilinear sparse matrix control systems, Annual Allerton Conference on Communication, Control, and Computing, Monticello, 2015, (**invited talk**)
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42. Behrouz Touri and *Bahman Ghahesifard*, Continuous-time distributed convex optimization on time-varying directed networks, Proceedings of the IEEE Conference on Decision and Control, Osaka, Japan, CA, 724-729, 2015 (**invited paper**)
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